Project Title	**CryptoCast: Multi-Horizon Bitcoin Price Forecasting Using Deep Learning**
Skills take away From This Project	• Time Series Analysis • Sequence Modeling • Deep Learning for Forecasting • Multi-Step Prediction • Model Comparison (CNN, RNN, LSTM, Transformer) • Evaluation of Financial Forecast Models • Real-World ML System Design
Domain	Financial Analytics / Cryptocurrency / Deep Learning

Problem Statement

Bitcoin prices are highly volatile and influenced by complex temporal patterns. Accurate short-term forecasting is crucial for traders, analysts, and automated trading systems. Traditional statistical methods struggle to capture non-linear dependencies and long-term temporal relationships present in crypto price movements.

This project focuses on building **deep learning models** that learn from **historical price sequences** to forecast Bitcoin prices over **multiple future horizons**.

Learners will design, train, and compare deep learning architectures to predict:

- **Next-day price (1D)**
- **3-day ahead price (3D)**
- **7-day ahead price (7D)**

Business Use Cases

- **Crypto Trading Platforms**: Short-term price forecasting signals
- **Algorithmic Trading Systems**: Multi-horizon decision making
- **Risk Management Tools**: Anticipate short-term volatility
- **Investment Analytics Dashboards**: Predictive insights
- **Educational Platforms**: Teaching real-world time-series DL

Dataset Description:

Link- [Dataset]

Dataset Columns

Column Name	Description

Date	Trading date
Price	Closing price (target variable)
Open	Opening price
High	Highest price of the day
Low	Lowest price of the day
Vol.	Trading volume
Change %	Daily percentage change

Project Objective

Develop deep learning models that:

1. Learn temporal patterns from **past 60 days of price data**
2. Forecast Bitcoin prices for **1-day, 3-day, and 7-day horizons**
3. Compare multiple deep learning architectures
4. Evaluate forecasting accuracy using appropriate metrics

Data Preparation Strategy

Sequence Generation

- Use **sliding window approach**
- Input sequence length: **60 past days**
- Output:
    - 1D forecast → next day price

- 3D forecast → next 3 days
- 7D forecast → next 7 days

Feature Selection

- Mandatory: `Price (Close)`
- Optional (Advanced):
  - Open, High, Low
  - Volume
  - Change %

Preprocessing Steps

- Sort data by date
- Handle missing values
- Scale features using **MinMaxScaler**
- Train-test split using **time-based split** (no shuffling)

Deep Learning Models to Implement

1. 1D Convolutional Neural Network (1D-CNN)

- Captures local temporal patterns
- Fast training
- Effective for short-term trends

2. Recurrent Neural Network (RNN)

- Sequential dependency modeling
- Baseline temporal model

3. Long Short-Term Memory (LSTM)

- Handles long-term dependencies
- Reduces vanishing gradient problem

4. Transformer (Time-Series Attention)

- Self-attention mechanism
- Captures global dependencies
- Advanced architecture for long sequences

Model Outputs

Each model must predict:

- **1-Day Price Forecast**
- **3-Day Price Forecast**
- **7-Day Price Forecast**

Approaches allowed:

- Separate models per horizon
- Single model with multi-output layer

Model Evaluation Metrics

Regression Metrics

- Mean Absolute Error (MAE)
- Root Mean Squared Error (RMSE)
- Mean Absolute Percentage Error (MAPE)

Visualization Metrics

- Actual vs Predicted price curves
- Error distribution plots
- Forecast horizon comparison

Expected Outcomes

- Functional multi-horizon forecasting system
- Comparative analysis of deep learning architectures
- Understanding of temporal dependency modeling
- Real-world financial forecasting exposure

Model Comparison Criteria

Model	Strength
1D-CNN	Short-term trend detection
RNN	Basic temporal learning

LSTM	Long-term dependency capture
Transformer	Global context awareness

Technical Tags

- Python
- Pandas
- NumPy
- TensorFlow / PyTorch
- Deep Learning
- Time Series Forecasting
- CNN / RNN / LSTM / Transformer

Project Deliverables

1. **Source Code**
    - Well-structured notebooks/scripts
    - Modular sequence generation code
2. **Project Report**
    - Problem understanding
    - Model architectures
    - Performance comparison
    - Business interpretation

3. **Visualizations**
    - Loss curves
    - Forecast plots
    - Horizon-wise comparison

Project Guidelines

Coding Standards:

- Clear function separation
- Reusable sequence generators
- Model configuration parameters

Validation Strategy:

- Time-series split (no random shuffle)
- Fixed random seeds
- Avoid data leakage

<u>References:</u>

In github, upload all your code files and submit the repository link in the Project section of your ZEN portal and you also have to appear for Project evaluation with one of our Mentors.

****If you don't know how to approach the project, kindly refer to the project orientation recording provided in this table.***

Project Live Evaluation	**Project Live Evaluation**

EDA Guide	Exploratory Data Analysis (EDA) G…
Capstone Explanation Guideline	Capstone Explanation Guideline
GitHub Reference	How to Use GitHub.pptx
Project_Series_English	Project_Series_English
Project_Series_Tamil	Project_Series_Tamil
Streamlit Doc	Streamlit doc
Streamlit Reference English	Streamlit Recordings
Streamlit Reference Tamil	Streamlit Recording Tamil
Project Orientation (English)	
Project Orientation (Tamil)	

PROJECT DOUBT CLARIFICATION SESSION (PROJECT AND CLASS DOUBTS)

About Session: The Project Doubt Clarification Session is a helpful resource for resolving questions and concerns about projects and class topics. It provides support in understanding project requirements, addressing code issues, and clarifying class concepts. The session aims to enhance comprehension and provide guidance to overcome challenges effectively.
Note: Book the slot at least before 12:00 Pm on the same day

Timing: Monday-Saturday (3:30PM to 4:30PM)

Session link :https://forms.gle/XC553oSbMJ2Gcfug9

LIVE EVALUATION SESSION (CAPSTONE AND FINAL PROJECT)

About Session: The Live Evaluation Session for Capstone and Final Projects allows participants to showcase their projects and receive real-time feedback for improvement. It assesses project quality and provides an opportunity for discussion and evaluation.
Note: This form will Open only on Saturday (after 2 PM) and Sunday on Every Week

Timing: Monday-Saturday (05:30PM to 07:00PM)

Booking link : https://forms.gle/1m2Gsro41fLtZurRA

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